Ahmedabad University

CSE400

CLIMATE AND WEATHER SYSTEMS

Problem Statement: *Probabilistic Precipitation Nowcasting*

M1 Submission

Section 2 - Group 2

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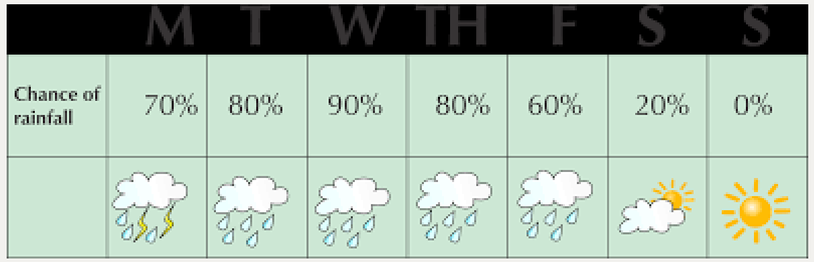
Probability Precipitation Nowcasting

- It is a short-term rainfall prediction (0-6 hours).
- It is used in disaster management.
- It is uncertain because of unpredictable weather changes.



	M	T	W	TH	F	S	S
Chance of rainfall	70%	80%	90%	80%	60%	20%	0%

Project Objective and Scope

- The objective of this project is to estimate the chances of rainfall based on the data provided by radar.
- The probability of rainfall is predicted for the next (0-6 hours).
- With the help of radar, multiple future possibilities of weather are generated.
- From these possibilities, chances of heavy rainfall can be derived.
- It can be useful in airport operations, flood warnings etc.

Project System

Radar Data → Motion Estimation → Extrapolation → Probabilistic adjustments → Ensemble Nowcasts

Main Components

- Rain shown by radar images.
- Method to track movement of rain, i.e, the speed and direction of rain.
- A model that predicts the probability of rain.

Data Flow

- Radar observations
- Multiple future possibilities of rain (forecasts)

Sources of Uncertainty

- Radar data is not always correct and may contain errors.
- Weather is uncertain and may change rapidly at any moment (rain can grow and change direction anytime.
- Assumptions are made in the model which for eg. rain moving at a constant speed.

Key Random Variables and their Roles

- **Rainfall Intensity (R) -** It represents precipitation rate (how heavy or light the rain is at that particular location).

- **Motion field (V) -** It describes rain cell movement (how rain clouds are moving, its speed and direction).

- **Noise terms (E) -** Unpredictable storm (shows unexpected changes in weather that cannot be predicted accurately).

Probability Models and Assumptions

Probability Models Used: STEPS (using PYSTEPS)

Assumptions

- A statistical pattern is followed by rainfall values.
- The clouds of rain move in the same direction for a short period of time.
- Random changes in rain are related across nearby areas.

Justification

- The above assumptions are made by observing real life rainfall data.

Probabilistic Reasoning and Inference Logic

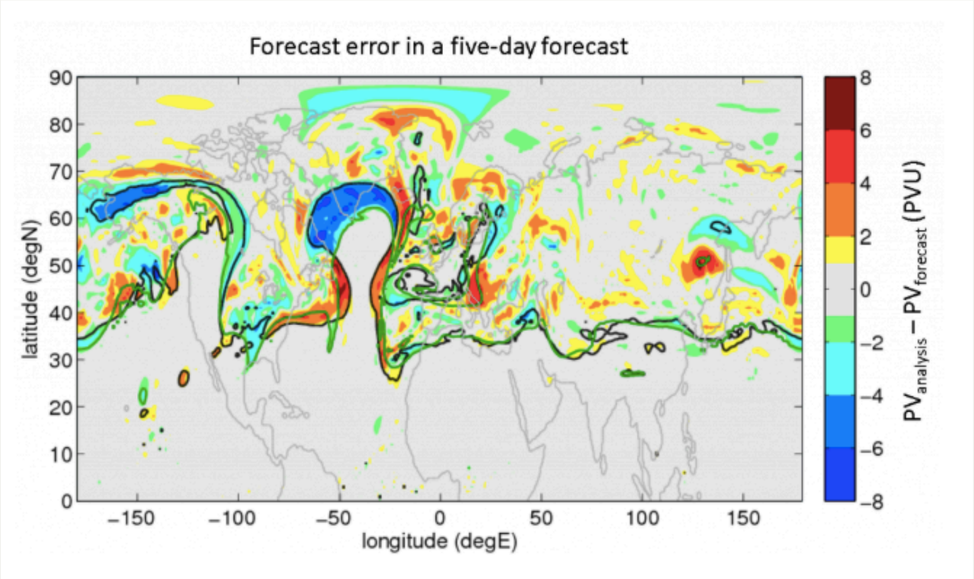
- The radar data include information about rainfall intensity and location, but noisy data are also considered.
- The rainfall field R(x,y,t) is an uncertain estimate of the actual state.
- The estimation of motion is uncertain because of errors in tracking.
- The uncertainty is propagated into the future as rainfall is forecasted.
- The stochastic components represent unknown growth, decay, and translation of rainfall.
- The multiple forecasts of rainfall show possible system behavior.
- The decisions are made based on probabilities of exceedance, not thresholds.

Operationalizing Probabilistic Reasoning

- The STEPS model is a rainfall extrapolation model that uses estimated motion fields.
- The precipitation distributions are used as input for the nowcasting system.
- Random perturbations are added to rainfall intensity fields.
- Gaussian noise fields add a controlled stochastic component.
- Motion uncertainty is represented by parallel and perpendicular perturbations.
- Each realization leads to a different rainfall outcome.
- An ensemble forecast is obtained by repeating the process.
- Ensemble outputs provide a summary of forecast uncertainty for 0-6 hours.

Current Limitations and Conceptual Gaps

- Limited ability to simulate the rapid onset or dissipation of storms.
- Simplified models of motion perturbations may not be able to simulate complex flow dynamics.
- Statistical properties assumed to be stationary for lead time.



- Gaussian noise may not be an appropriate model for real precipitation variability.
- Finite ensemble size limits ability to simulate extremes.
- Correlation between motion and intensity uncertainty is not modeled.
- Probabilistic simulation is dependent on external thresholds.

Planned Refinements and Role Coordination

- More realistic representation of motion uncertainties rather than fixed directional errors.
- Improved stochastic intensity representation for convective cases.
- New ensemble calibration using verification standards.
- Reexamination of assumptions about stationarity and Gaussian processes.
- Extension of ensemble size when computationally feasible.

Role Coordination:

- Shine and Dhruvi - Video/PPT
- Rim and Sharanya - Scribe
- Vamsi - Concept Map

Summary of Group-Level Understanding

- Nowcasting of rainfall necessarily involves uncertainty in short-term forecasting.
- Deterministic forecasting is insufficient to represent variability of the future.
- Probabilistic inference allows representation of uncertainty.
- STEPS translates radar data into ensembles of possible rainfall outcomes.
- Stochastic perturbations represent uncertain physical processes.
- Ensemble forecasting allows representation of risk as a probability.
- There is a consensus in the group that models should be able to represent uncertainty and be interpretable.
- Future work will aim to incorporate realism while maintaining tractability.

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THANK YOU!